The internal resistance of the battery,

$$r = \left[\frac{\varepsilon - V}{V}\right] R = \left[\frac{12 - 11.79}{11.79}\right] \times 3 = 0.05\ \Omega$$

(b) The power delivered by the battery $P = I\varepsilon = 3.93 \times 12 = 47.1$ W

The power delivered to the resistor = I^2R = 46.3 W

The remaining power P = (47.1 – 46.3) = 0.8 W is delivered to the internal resistance and cannot be used to do useful work. (It is equal to I^2r).

2.4.3 Cells in series

Several cells can be connected to form a battery. In series connection, the negative terminal of one cell is connected to the positive terminal of the second cell, the negative terminal of second cell is connected to the positive terminal of the third cell and so on. The free positive terminal of the first cell and the free negative terminal of the last cell become the terminals of the battery.

Suppose n cells, each of emf ε volts and internal resistance r ohms are connected in series with an external resistance R as shown in Figure 2.21

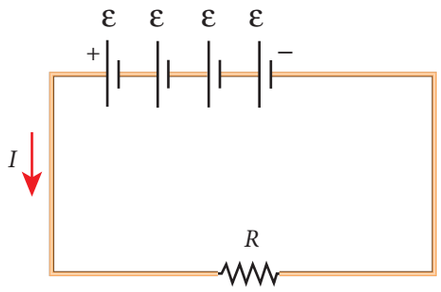


Figure 2.21 cells in series

The total emf of the battery = $n\varepsilon$

The total resistance in the circuit = $nr + R$

By Ohm's law, the current in the circuit is

$$I = \frac{\text{total emf}}{\text{total resistance}} = \frac{n\varepsilon}{nr + R} \qquad (2.39)$$

Case (a) If $r << R$, then,

$$I = \frac{n\varepsilon}{R} \approx nI_1 \qquad (2.40)$$

where, I_1 is the current due to a single cell $\left(I_1 = \frac{\varepsilon}{R}\right)$

Thus, if r is negligible when compared to R the current supplied by the battery is n times that supplied by a single cell.

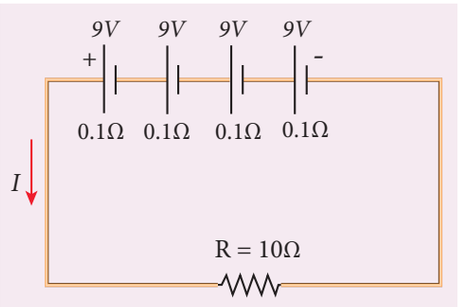
Case (b) If $r >> R$, $I = \frac{n\varepsilon}{nr} \approx \frac{\varepsilon}{r}$ (2.41)

It is the current due to a single cell. That is, current due to the whole battery is the same as that due to a single cell and hence there is no advantage in connecting several cells.

Thus series connection of cells is advantageous only when the effective internal resistance of the cells is negligibly small compared with R.

EXAMPLE 2.18

From the given circuit,



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